

Managing User Passwords and Authentication in Linux



Andrew Mallett

Author and Trainer

@theurbanpenguin www.theurbanpenguin.com



Overview



Linux Passwords and Authentication

- `/etc/shadow` and `getent shadow`
- `man 5 shadow`
- Setting passwords with `passwd` and `chpasswd`
- Splitting the password field
- Authenticating users
- Group passwords and `/etc/gshadow`
- Self-service access to resources
- Adding group members and administrators via the `gshadow` file using `gpasswd`

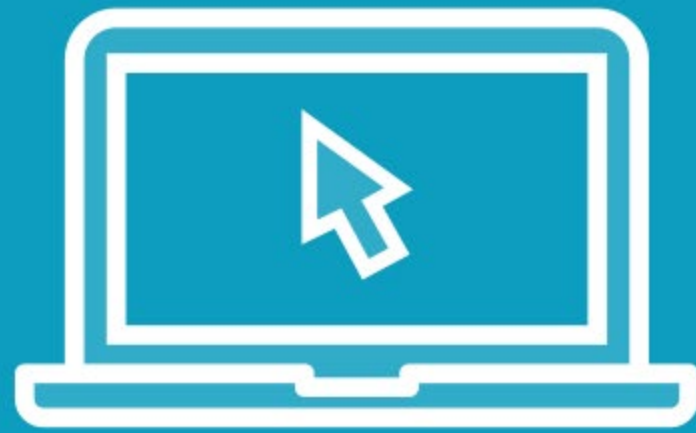


```
$ ls -l /etc/shadow
$ ls -l /usr/bin/passwd
$ sudo cat /etc/shadow
$ sudo getent shadow vagrant
$ man 5 shadow
```

User Passwords

Using the **adduser** Perl script in Ubuntu, you will be prompted for a user password. **Useradd** has an option **-p** for the password but expects an encrypted password (password hash). Passwords are stored in the file **/etc/shadow**, a password starting with an exclamation mark is invalid.

Demo



Investigating User Passwords

- New user with useradd
- Checking the user password
- Password last changed
- Aging information withchage



Manage User Passwords

passwd

Users can change their own password or administrators change user passwords interactively

chpasswd

Administrators can set passwords for many users and non-interactively making great scripting

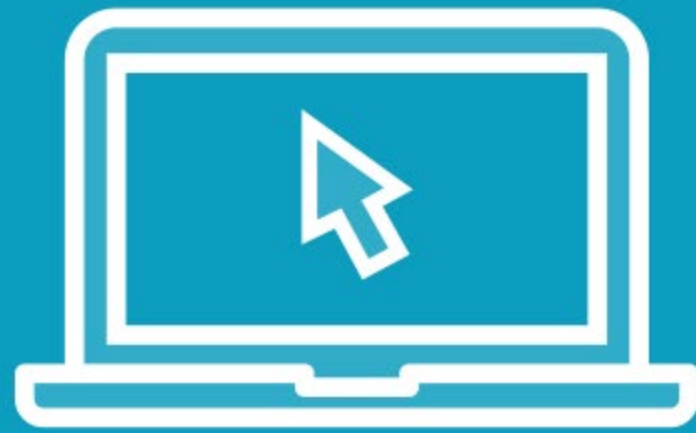



```
$ sudo passwd user1
$ echo 'user1:[Password123?]' | sudo chpasswd
$ sudo chpasswd
user1:[Password1]
user2:[Password2]
CTRL+D
$ sudo chpasswd < users
```

Managing User Passwords

Although we are used to the **passwd** command, we have to enter a user's password twice to set it. If we are creating many users, it is not the best idea. To script password management we have **chpasswd**.

Demo



Managing User Passwords:

- passwd
- chpasswd

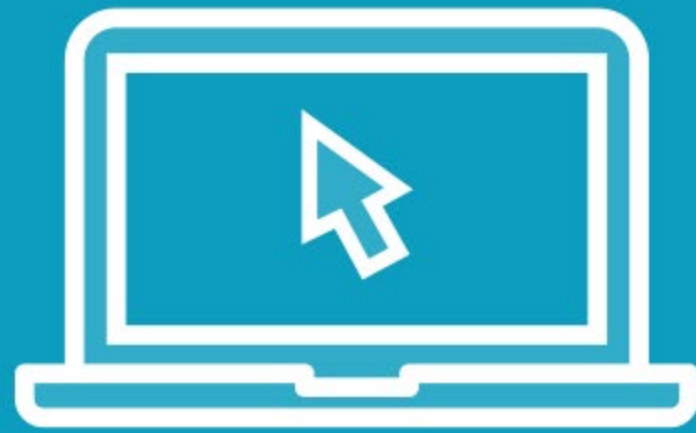


```
$ sudo getent shadow $USER | cut -f2 -d: | \
  awk -F'$' '{ print "ALG:\t" $2 "\nSALT:\t" $3 "\nHASH:\t" $4 }'
```

Splitting the Password Field

The user password is the second field in the `/etc/shadow` file; however, there are three entities to a password. The algorithm, the salt and the password hash. Using AWK, we can spilt the password to understand the elements

Demo



Splitting the ~~ATOM~~ Password:

- Using AWK
- Algorithm
- Salt
- Hash

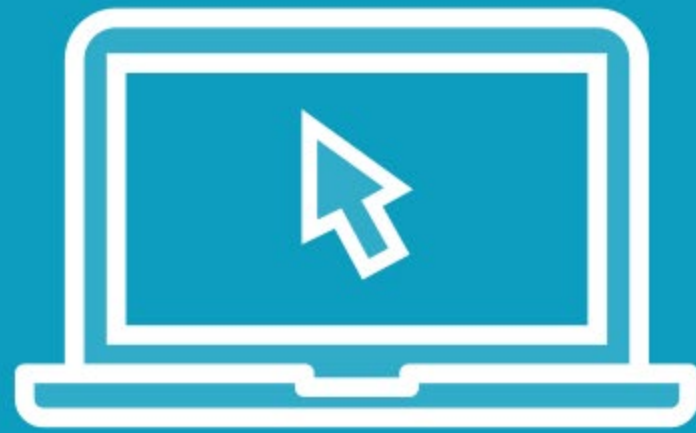


```
$ openssl passwd -6 Password1  
$ openssl passwd -6 -salt XN2wmxkkdjLiIOUk Password1
```

Authentication

The user password is stored as a one-way hash meaning that even the root user cannot tell you what your password is. Authentication occurs when the entered user password produces the same hash when combined with the same algorithm and same salt value. Neither the salt or algorithm is encrypted.

Demo



Understanding User Authentication:

- openssl passwd module
- generating passwords
- authenticating users



Group Passwords



Passwords can be set on groups but will make the group **LESS** secure and **NOT** more secure. If a group has a password any user can switch to the group if they know the password. This gives self-service access but is not ideal.

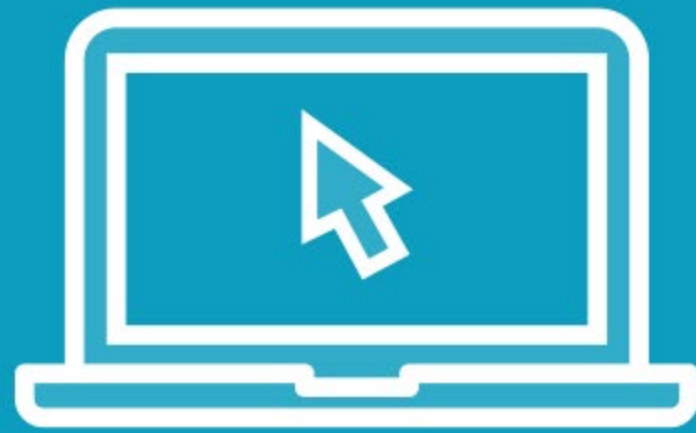


```
$ sudo groupadd helpdesk  
$ sudo gpasswd -a vagrant helpdesk  
$ sudo gpasswd -A vagrant helpdesk  
  
$ gpasswd -a ubuntu helpdesk
```

Managing Groups

The `/etc/gshadow` file not only stores the group password, if set. It can store group administrators. An administrator can manage group membership with the `gpasswd` command.

Demo



Managing User Groups Membership:

- gpsswd
- group admins



Summary



User and Group Passwords

- /etc/shadow
- /etc/gshadow
- \$alg\$salt\$hash
- openssl passwd
- group passwords
- group administrators



Managing System-Wide Login Scripts

